Table S4. Additional module validation panels

| Function | Independent reference | Observed result | Scope limit |
|---|---|---|---|
| Promoter extraction | Direct 1-based inclusive FASTA slicing and reverse complement; bedtools 2.31.1 | 8/8 exact sequences, 2,200 bp each, both strands represented | Eight loci on one Arabidopsis chromosome |
| CDS sequence quality | Independent base-count/frame/start/stop oracle | Four RBCS CDS models (411, 561, 546, 546 nt) matched; no ambiguous bases flagged | Four models from one family |
| Protein similarity | Test-side Smith–Waterman and BLASTP 2.12.0+ | All 16 within-panel alignment metric checks; 12/12 non-self RBCS and 8/8 negative pair classifications agree; 20/20 in the cross-species mature-RBCS panel | Small one-family smoke test; not orthology, sensitivity, or specificity evaluation |
| Expression grouping | Expression Atlas E-MTAB-7978 source values | Selected means/medians reproduced; measured zeros retained | Single-assay group arithmetic, not differential expression |
| Domain mapping | Independent Python reconstruction | 16 domain/deletion coordinate records across 13 genes matched | Existing annotated models; not a curated multi-species accuracy set |
| Project persistence | JavaFX regression actions | Save, replace, restore, locus state and outputs verified in CI | Software behavior, not usability study |